Exercise 2G

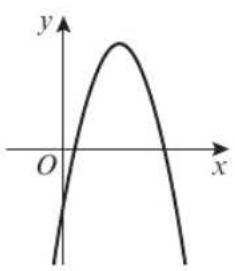
1 a Calculate the value of the discriminant for each of these five functions:

i $f(x) = x^2 + 8x + 3$ **ii** $g(x) = 2x^2 - 3x + 4$ **iii** $h(x) = -x^2 + 7x - 3$

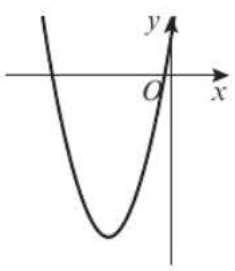
iv $j(x) = x^2 - 8x + 16$ **v** $k(x) = 2x - 3x^2 - 4$

b Using your answers to part **a**, match the same five functions to these sketch graphs.

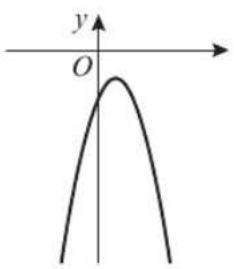
i



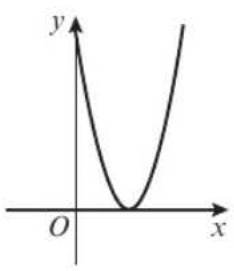
ii



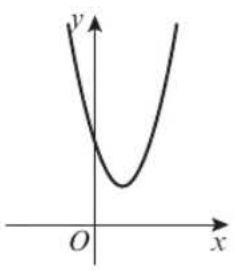
iii



iv



v



E/P **2** Find the values of k for which $x^2 + 6x + k = 0$ has two real solutions. **(2 marks)**

E/P **3** Find the value of t for which $2x^2 - 3x + t = 0$ has exactly one solution. **(2 marks)**

E/P **4** Given that the function $f(x) = sx^2 + 8x + s$ has equal roots, find the value of the positive constant s. **(2 marks)**

E/P **5** Find the range of values of k for which $3x^2 - 4x + k = 0$ has no real solutions. **(2 marks)**

E/P **6** The function $g(x) = x^2 + 3px + (14p - 3)$, where p is an integer, has two equal roots.

a Find the value of p. **(2 marks)**

b For this value of p, solve the equation $x^2 + 3px + (14p - 3) = 0$. **(2 marks)**

E/P **7** $h(x) = 2x^2 + (k + 4)x + k$, where k is a real constant.

a Find the discriminant of $h(x)$ in terms of k. **(3 marks)**

b Hence or otherwise, prove that $h(x)$ has two distinct real roots for all values of k. **(3 marks)**

Problem-solving

If a question part says 'hence or otherwise' it is usually easier to use your answer to the previous question part.

Challenge

a Prove that, if the values of a and c are given and non-zero, it is always possible to choose a value of b so that $f(x) = ax^2 + bx + c$ has distinct real roots.

b Is it always possible to choose a value of b so that $f(x)$ has equal roots? Explain your answer.

2.6 Modelling with quadratics

A **mathematical model** is a mathematical description of a real-life situation. Mathematical models use the language and tools of mathematics to represent and explore real-life patterns and relationships, and to predict what is going to happen next.

Models can be simple or complicated, and their results can be approximate or exact. Sometimes a model is only valid under certain circumstances, or for a limited range of inputs. You will learn more about how models involve simplifications and assumptions in Statistics and Mechanics.

Quadratic functions can be used to model and explore a range of practical contexts, including projectile motion.

Example 15

A spear is thrown over level ground from the top of a tower.
The height, in metres, of the spear above the ground after t seconds is modelled by the function:

$$h(t) = 12.25 + 14.7t - 4.9t^2,\ t \geqslant 0$$

a Interpret the meaning of the constant term 12.25 in the model.

b After how many seconds does the spear hit the ground?

c Write $h(t)$ in the form $A - B(t - C)^2$, where A, B and C are constants to be found.

d Using your answer to part **c** or otherwise, find the maximum height of the spear above the ground, and the time at which this maximum height is reached.

a The tower is 12.25 m tall, since this is the height at time 0.

b When the spear hits the ground, the height is equal to 0.

$$12.25 + 14.7t - 4.9t^2 = 0$$

Using the formula, where $a = -4.9$, $b = 14.7$ and $c = 12.25$,

$$t = \frac{-14.7 \pm \sqrt{14.7^2 - 4(-4.9)(12.25)}}{(2 \times -4.9)}$$

$$t = \frac{-14.7 \pm \sqrt{456.19}}{-9.8}$$

$t = -0.679$ or $t = 3.68$ (to 3 s.f.)

As $t \geqslant 0$, $t = 3.68$ seconds (to 3 s.f.).

c $12.25 + 14.7t - 4.9t^2$

$= -4.9(t^2 - 3t) + 12.25$

$= -4.9((t - 1.5)^2 - 2.25) + 12.25$

$= -4.9((t - 1.5)^2 + 11.025 + 12.25)$

$= 23.275 - 4.9(t - 1.5)^2$

So $A = 23.275$, $B = 4.9$ and $C = 1.5$.

d The maximum height of the spear is 23.275 metres, 1.5 seconds after the spear is thrown.

Problem-solving

Read the question carefully to work out the meaning of the constant term in the **context of the model**. Here, $t = 0$ is the time the spear is thrown.

To solve a quadratic, factorise, use the quadratic formula, or complete the square.

Give any non-exact numerical answers correct to 3 significant figures unless specified otherwise.

Always interpret your answers in the context of the model. t is the time after the spear was thrown so it must be positive.

$4.9(t - 1.5)^2$ must be positive or 0, so $h(t) \leqslant 23.275$ for all possible values of t.

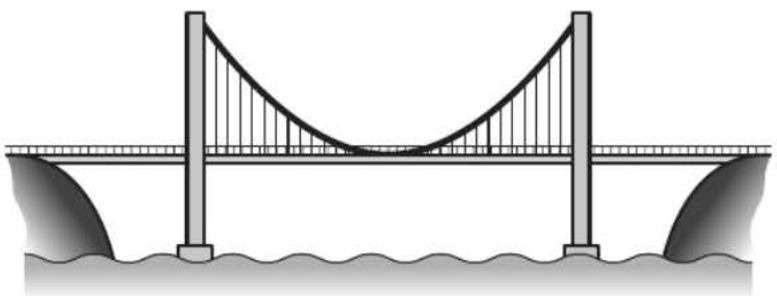
The turning point of the graph of this function would be at (1.5, 23.275). You may find it helpful to draw a sketch of the function when working through modelling questions.

Online Explore the trajectory of the spear using GeoGebra.



Exercise 2H

E/P 1 The diagram shows a section of a suspension bridge carrying a road over water.



Problem-solving

For part **a**, make sure your answer is in the context of the model.

The height of the cables above water level in metres can be modelled by the function $h(x) = 0.000\,12x^2 + 200$, where x is the displacement in metres from the centre of the bridge.

a Interpret the meaning of the constant term 200 in the model. **(1 mark)**

b Use the model to find the two values of x at which the height is 346 m. **(3 marks)**

c Given that the towers at each end are 346 m tall, use your answer to part **b** to calculate the length of the bridge to the nearest metre. **(1 mark)**

E/P 2 A car manufacturer uses a model to predict the fuel consumption, y miles per gallon (mpg), for a specific model of car travelling at a speed of x mph.

$$y = -0.01x^2 + 0.975x + 16,\ x > 0$$

a Use the model to find two speeds at which the car has a fuel consumption of 32.5 mpg. **(3 marks)**

b Rewrite y in the form $A - B(x - C)^2$, where A, B and C are constants to be found. **(3 marks)**

c Using your answer to part **b**, find the speed at which the car has the greatest fuel efficiency. **(1 mark)**

d Use the model to calculate the fuel consumption of a car travelling at 120 mph. Comment on the validity of using this model for very high speeds. **(2 marks)**

E/P 3 A fertiliser company uses a model to determine how the amount of fertiliser used, f kilograms per hectare, affects the grain yield g, measured in tonnes per hectare.

$$g = 6 + 0.03f - 0.000\,06f^2$$

a According to the model, how much grain would each hectare yield without any fertiliser? **(1 mark)**

b One farmer currently uses 20 kilograms of fertiliser per hectare. How much more fertiliser would he need to use to increase his grain yield by 1 tonne per hectare? **(4 marks)**

E/P 4 A football stadium has 25 000 seats. The football club know from past experience that they will sell only 10 000 tickets if each ticket costs £30. They also expect to sell 1000 more tickets every time the price goes down by £1.

a The number of tickets sold t can be modelled by the linear equation $t = M - 1000p$, where £p is the price of each ticket and M is a constant. Find the value of M. **(1 mark)**

The total revenue, £r, can be calculated by multiplying the number of tickets sold by the price of each ticket. This can be written as $r = p(M - 1000p)$.

b Rearrange r into the form $A - B(p - C)^2$, where A, B and C are constants to be found. **(3 marks)**

c Using your answer to part **b** or otherwise, work out how much the football club should charge for each ticket if they want to make the maximum amount of money. **(2 marks)**

Challenge

Accident investigators are studying the stopping distance of a particular car.
When the car is travelling at 20 mph, its stopping distance is 6 feet.
When the car is travelling at 30 mph, its stopping distance is 14 feet.
When the car is travelling at 40 mph, its stopping distance is 24 feet.

The investigators suggest that the stopping distance in feet, d, is a quadratic function of the speed in miles per hour, s.

a Given that $\mathrm{d}(s) = as^2 + bs + c$, find the values of the constants a, b and c.

b At an accident scene a car has left behind a skid that is 20 feet long.
Use your model to calculate the speed that this car was going at before the accident.

Hint Start by setting up three simultaneous equations. Combine two different pairs of equations to eliminate c. Use the results to find the values of a and b first.

Mixed exercise 2

1 Solve the following equations without a calculator. Leave your answers in surd form where necessary.

a $y^2 + 3y + 2 = 0$ **b** $3x^2 + 13x - 10 = 0$ **c** $5x^2 - 10x = 4x + 3$ **d** $(2x - 5)^2 = 7$

2 Sketch graphs of the following equations:

a $y = x^2 + 5x + 4$ **b** $y = 2x^2 + x - 3$ **c** $y = 6 - 10x - 4x^2$ **d** $y = 15x - 2x^2$

E **3** $\mathrm{f}(x) = x^2 + 3x - 5$ and $\mathrm{g}(x) = 4x + k$, where k is a constant.

a Given that $\mathrm{f}(3) = \mathrm{g}(3)$, find the value of k. **(3 marks)**

b Find the values of x for which $\mathrm{f}(x) = \mathrm{g}(x)$. **(3 marks)**

4 Solve the following equations, giving your answers correct to 3 significant figures:

a $k^2 + 11k - 1 = 0$ **b** $2t^2 - 5t + 1 = 0$ **c** $10 - x - x^2 = 7$ **d** $(3x - 1)^2 = 3 - x^2$

5 Write each of these expressions in the form $p(x + q)^2 + r$, where p, q and r are constants to be found:

a $x^2 + 12x - 9$ **b** $5x^2 - 40x + 13$ **c** $8x - 2x^2$ **d** $3x^2 - (x + 1)^2$

E **6** Find the value k for which the equation $5x^2 - 2x + k = 0$ has exactly one solution. **(2 marks)**